Google Satellite Embedding per small area (2017-2024) data product

This technical report aims to describe the access, processing and development of the Imago Embedding data product, which is based on AlphaEarth Foundations Satellite Embedding Dataset V1 available in Google Earth Engine for 2017-2024.

This data product provides Embedding values (64-dimensional, L2-normalised vectors, with values ranging between −1 and +1) at 10 meter spatial resolution, aggregated to small-area geographies — Lower Layer Super Output Areas (LSOAs).

> *Although 2017 featured data quality issues (see here), they have been addressed in the latest version of the > original Embedding data collection available on Google Earth Engine.*
> *From the AI for Good-2026 Workshop: The issue was related to dropout of some Sentinel-1 images, but the overall impact on dataset accuracy was quite > low. As a rule of thumb, **Google Earth Engine data is always a source of truth** and always contains the latest and updated version.*

For further details on the original Embedding dataset, see this blogpost.

The developed workflow involves two pipelines:

1. The export pipeline uses Google Earth Engine (GEE) to export the 20x20 KM tiles.

2. Aggregation of embeddings at the level of small geographies (LSOA).

Data specification

Throughout the IMAGO workflow, data changes its form twice, resulting in three data stages:
0. Input data:

- GeoTIFF format, optionally with COG layout (cannot be parameterised).
- 64 bands (from A00 to A63).
- Int16 data type, LZW compression.
- CRS - local UTM zone projection; spatial resolution 10 m.
- NoData is undefined by default (can be defined).

- Usually features one extra pixel along northern and eastern tile edges (so if expected 2000 x 2000 pixels, the output will contain 2001 x 2001 pixels).

1. Intermediate data:

- GeoTIFF format without COG layout.
- 64 bands (from A00 to A63).
- Int16 data type, LZW compression.
- CRS - EPSG:27700, OSGB36/British National Grid; spatial resolution 10 m.
- NoData is undefined with one extra pixel along the northern and eastern tile edges. In the further processing pixels equal to 0 are filtered out to avoid calculation distortions.
- Gridded by 20 x 20 km National Grid tiles.
- Value magnitude ranges from -32767 to +32767 to reduce the output size in average by one third of the original size. To work with original decimal values (to 'descale') which range from -1 to +1, the following formula can be used: $\mathrm{x_{original}} = \frac{\mathrm{x_{int16}}}{32767}$
- The scaled `int16` dataset preserves the original `float64` values with very high precision (*Figure 1*). The mean error and RMSE are on the 6th decimal place, while the maximum absolute error occurs at the 5th decimal place (validated for SJ26 tile, 2024).

Tile	Band	Max_err	Mean_err	RMSE
SJ26-2024	1	0.000015	0.000008	0.000009
SJ26-2024	2	0.000015	0.000007	0.000009
SJ26-2024	3	0.000015	0.000008	0.000009
SJ26-2024	4	0.000015	0.000007	0.000008
SJ26-2024	5	0.000015	0.000007	0.000008
SJ26-2024	6	0.000015	0.000008	0.000009
SJ26-2024	7	0.000015	0.000007	0.000008
SJ26-2024	8	0.000015	0.000007	0.000008
SJ26-2024	9	0.000015	0.000007	0.000008
SJ26-2024	10	0.000015	0.000007	0.000008
SJ26-2024	11	0.000015	0.000007	0.000008

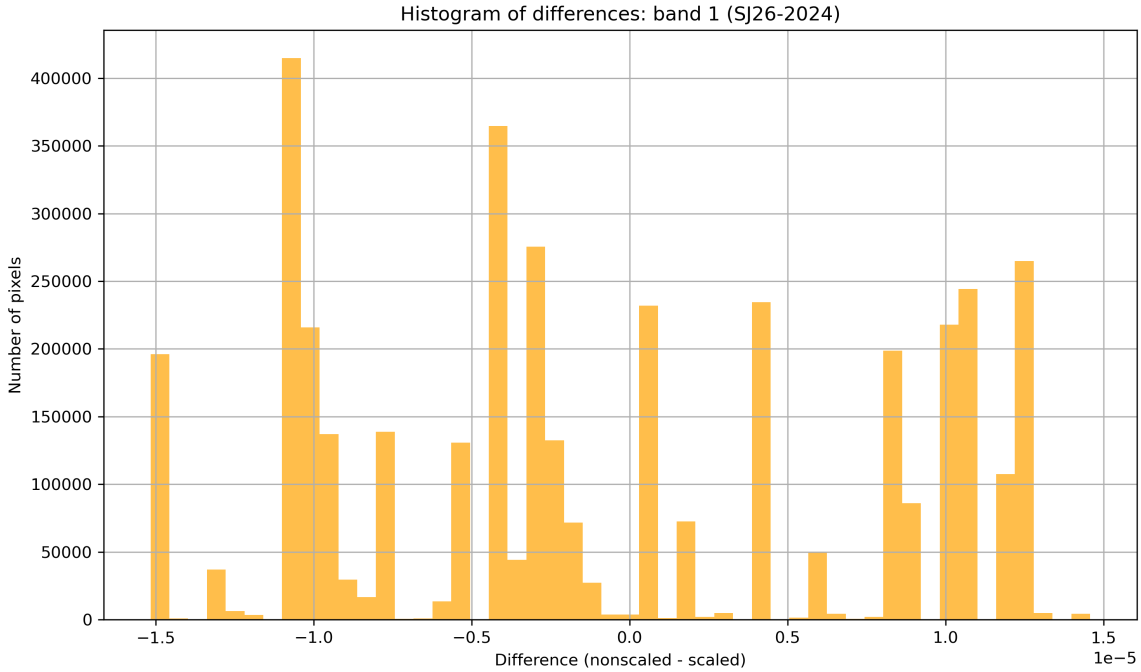

Figure 1: Scaling error statistics snippet (IMAGO) for maximum error, mean error and MSE (above) and difference between the nonscaled and scaled values, A00 dimension, 2024, Tile SJ26 (below)

Google Embedding dataset by Source Cooperative organised in a different way. It is stored in signed 8-bit data type through nonlinear scaling, with `-128` as a nodata value and the spatial index recorded within the filenames. This also allows to considerably reduce the filesize (3.45 GB for 8192 x 8192 pixel image), but has a lower accuracy, especially regarding the maximum error (*Figure 2*).

Tile	Band	Max_err	Mean_err	RMSE
S084-2024	1	0.090058	0.000044	0.000898
S084-2024	2	0.104760	0.000053	0.001083
S084-2024	3	0.121553	0.000062	0.001264
S084-2024	4	0.109496	0.000054	0.001218
S084-2024	5	0.077017	0.000035	0.000713
S084-2024	6	0.142714	0.000069	0.001510
S084-2024	7	0.091719	0.000043	0.000843
S084-2024	8	0.139208	0.000055	0.001203
S084-2024	9	0.097686	0.000046	0.000939
S084-2024	10	0.113495	0.000054	0.001056
S084-2024	11	0.120692	0.000061	0.001218

Figure 2: Scaling error statistics snippet (Source Cooperative) for maximum error, mean error and MSE

2. **Output data:**

- GeoPackage format.
- Built on the LSOA features.
- Includes additional 64 columns for each statistical parameter (mean, median, min, max, std, sum, count), which represent Embedding values for each dimension.
- Magnitude of embedding value ranges from -1 to +1 (accordingly to input data).
- Embedding columns have floating-point (double-precision) data types, except from count (integer).

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1. Google Earth Engine pipeline

A command line tool to access annual satellite collections, filter by the area and timeframe of interest, and export in GeoTIFF format to Google Drive or Google Cloud Storage has been developed.

Main requirements are:

- Google Earth Engine account and created project (works for projects of any level)
- Area of interest, which can be tiled (for example, the UK gridded by tiles of 20km x 20 km)
- Year of interest

The pipeline utilises a simple processing and I/O structure by accesssing data from Google Earth Engine, checking its match with the area and timeframe of interest, and exporting imagery batches through Google Earth Engine API (*Figure 3*).

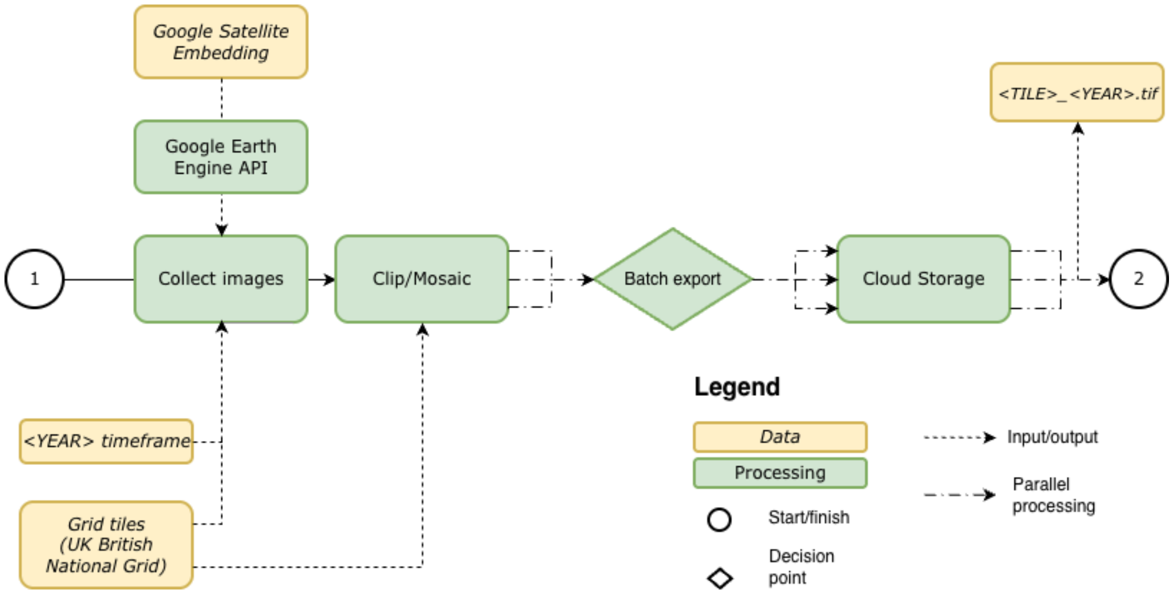

Figure 3: Conceptual flowchart, illustrating the input data, input/output operations, processing steps and output data in the Google Earth Engine download pipeline

Any annual collections can be used as input for further export, for example, ESA WorldCover 10m v100 or ESA WorldCereal Active Cropland 10 m v100. Dataset with higher granularity will not be correctly aggregated by this tool.

To test the tool and explore outputs it can be recommended to consider **the Greater London case study area**, which interesects 11 tiles of 20km x 20 km (according to the used LSOA polygons and London Datastore). However, almost 98% of the Greater London area is covered by six tiles: TQ06, TQ08, TQ26, TQ28, TQ46, TQ48. This area has been used to demonstrate the data product in the hands-on workshop introducing satellite (geo-)embeddings and their practical applications, co-organised by Imago and MHCLG.

Command-line tool (CLI)

To run the tool in default mode, use:
`python src/download_ee.py`

Additional help and usage examples can be explored by: `python src/download_ee.py --help`

Positional arguments are not used, as input names are long and non-obvious without a name. Therefore, the CLI tool uses `option` only.

The following options are provided:

Option	Description	Default
`--collection`	Earth Engine image collection to extract	GOOGLE/SATELLITE_EMBEDDING/V1/ANNUAL
`--project` , `-p`	Google Earth Engine project ID	imago
`--auth-mode`	Earth Engine authentication mode	gcloud
`--tiles`	Path to the gridded area of interest (tiles)	data/uk_1km_grid_sample1.gpkg
`--storage`	Export destination – either Google Cloud	cloud

	Storage or Google Drive	
--folder	Cloud Storage bucket or Google Drive folder	embed2social-storage
--year , -y	Year to extract dataset for	2024
--verbose , -v	Enable verbose logging (WARNING: can produce very large logs)	False
--cog	Export as Cloud Optimized GeoTIFF (COG)	False
--crs	Coordinate Reference System (CRS) of the output	EPSG:27700
--res	Spatial resolution of output	10
--scale	Scale to Int16, multiplying by 32767	False

Tiling

Imago intermediate dataset in GeoTIFF format is gridded by 20 x 20 km British National Grid tiles, which accounts to 858 tiles in total for the UK.

Aside from original Google Satellite Embedding, there exist a Source Cooperative data product. AlphaEarth Foundation Embeddings in this product are internally tiled - the boundaries of tiles can be found here in `aef.index` file (incl. GeoPackage). UK is covered by 39 Embedding tiles in the original dataset, which generally follow the outlines of UTM zones.

However, testing with GEE export in verbose mode revealed that the original Embedding tiling differs from the published Source Cooperative tile boundaries. For example, the area of interest which intersects four Source Cooperative tiles, overlays only two original Google Satellite Embedding images while the outputs consistent and covers the whole area of interest.

Performance

There exist different time concepts when talking about cloud computing.

1. Python **pipeline runtime** (client wall-clock time)
   It's time on client machine between two Python timestamps, so it doesn't indicate billing/quotas.
2. **EECU (Earth Engine Compute Units)**
   Computes consumption across all parallel workers (depends on CPU/memory) and shows billing/quotas.
   Includes I/O reads, arrays manipulations, but doesn't include latency/queue scheduling/client waiting/Google Drive or Cloud uploads.
3. Task runtime
   Includes queue wait time, server execution time, writing I/O (doesn't include time before scheduling a task)

Python runtime is a sum of task runtimes, client-side preparation, initialization and scheduling time before tasks start to run on Earth Engine.

Even the same requests might be processed in a very different time (see here). It has been found that EECU time for the same area of interest can vary by a factor of 2.8, while total runtime can vary by up to a factor of 8.8. However, the pipeline runtime and billable EECU time can be roughly predicted for large extracts (see below).

Pipeline runtime and data volume

By tiling, compressing, and scaling the embedding data, the total data size was reduced from an estimated ~4 TiB to approximately 2 TiB.

Year	Pipeline runtime (s)	Pipeline runtime (h)	EECU time (s)	EECU time (h)	Total size (Gb)
2024	27,420.6247	7.6168	278,886.6827	77.4685	258.24
2023	30,263.8161	8.4066	264,226.6384	73.3962	258.64
2022	25,515.9630	7.0877	257,439.4404	71.5109	258.64
2021	27,541.3252	7.6504	262,592.8037	72.9424	258.48
2020	24,354.6962	6.7652	275,837.1997	76.6214	258.76
2019	41,363.0000	11.489	302,880.4143	84.1334	258.11
2018	32,771.5112	9.1032	262,321.2517	72.867	258.44

2017	42,716.2087	11.8656	287,451.6494	79.8477	260.33
AVERAGE	31,493.3931	8.7482	273,954.51	76.0985	258.705
TOTAL	251,947.145	**69.9853**	2,191 636.08	**608.7877**	**2.02 TiB**

> *Batch export*
>
> *In this pipeline, batch export tasks are used: `Export.image.toDrive` or `Export.image.toCloudStorage` which provide identical outputs:*
>
> - *The main parameters of these functions are the same*
> - *Only two formats available: GeoTIFF and TFR (tensorflowrecords): https://towardsdatascience.com/tfrecords-explained-24b8f2133282/ - serialized JSONs to binary sequences*
> - *LZW compression always applied to TIF, regardless whether it's COG or not*
> - *`maxPixels` parameter is crucial - if exceeded, raises Error*
> - *`shardSize` doesn't change metadata, but might contribute to higher EECU usage (for 101*101 pixel image shardsize=55 increases EECU time roughly by factor of 2). It's a sort of internal tiling, which can be used to reduce memory per worker and avoid errors (safer not means faster). Introduces overhead for small exports though.*
>
> *Execution caveats:*
>
> - *The number of concurrently running tasks vary depending on the project configuration. For the non-commercial project, it's up to three-four, for the professional paid project - 20.*
> - *Maximum number of concurrently running tasks is not guaranteed (eg, for non-commercial projects one-task cap has been experienced).*
> - *It is not possible to exactly predict computation performance, as computational capacity is volatile and depends on caching/EE algorithm changes/libraries changes, aside from different underlying data.*
> - *Number of workers is determined by EE service configuration/ability to parallelise the job, and is **NOT CONFIGURABLE**.*

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2. LSOA aggregation

The LSOA aggregation pipeline takes the tiled GeoTIFF outputs from the previous step, and aggregates them to LSOA boundaries.
The output is a GeoPackage file with one row per LSOA, and columns for each of the 64

embedding dimensions, with values representing the mean embedding value across all pixels that fall within the LSOA boundary.

This LSOA extraction step can be run for all years in the dataset, using `bash` scripting and configuration file templates. These configuration
files differ only in the year of interest - the input/output paths and other relevant parameters are amended purely using the year of interest.

Figure 4 below shows the LSOA extraction workflow. The pipeline (at step 1 in the flowchart) is controlled by a script `run_all_years.sh`.
This script generates a configuration file (`config_$YEAR.yaml`) and a separate runscript (`run_$YEAR.sh`) for each year
of interest (default 2017-2024) based on these templates. The configuration file contains all relevant parameters for the LSOA extraction step,
including input/output paths, and other parameters such as the number of workers to use, and the amount of memory to give to each worker.
The runscript then runs the LSOA extraction pipeline for that year (`embed_to_lsoa.py`), using the generated configuration file as input.

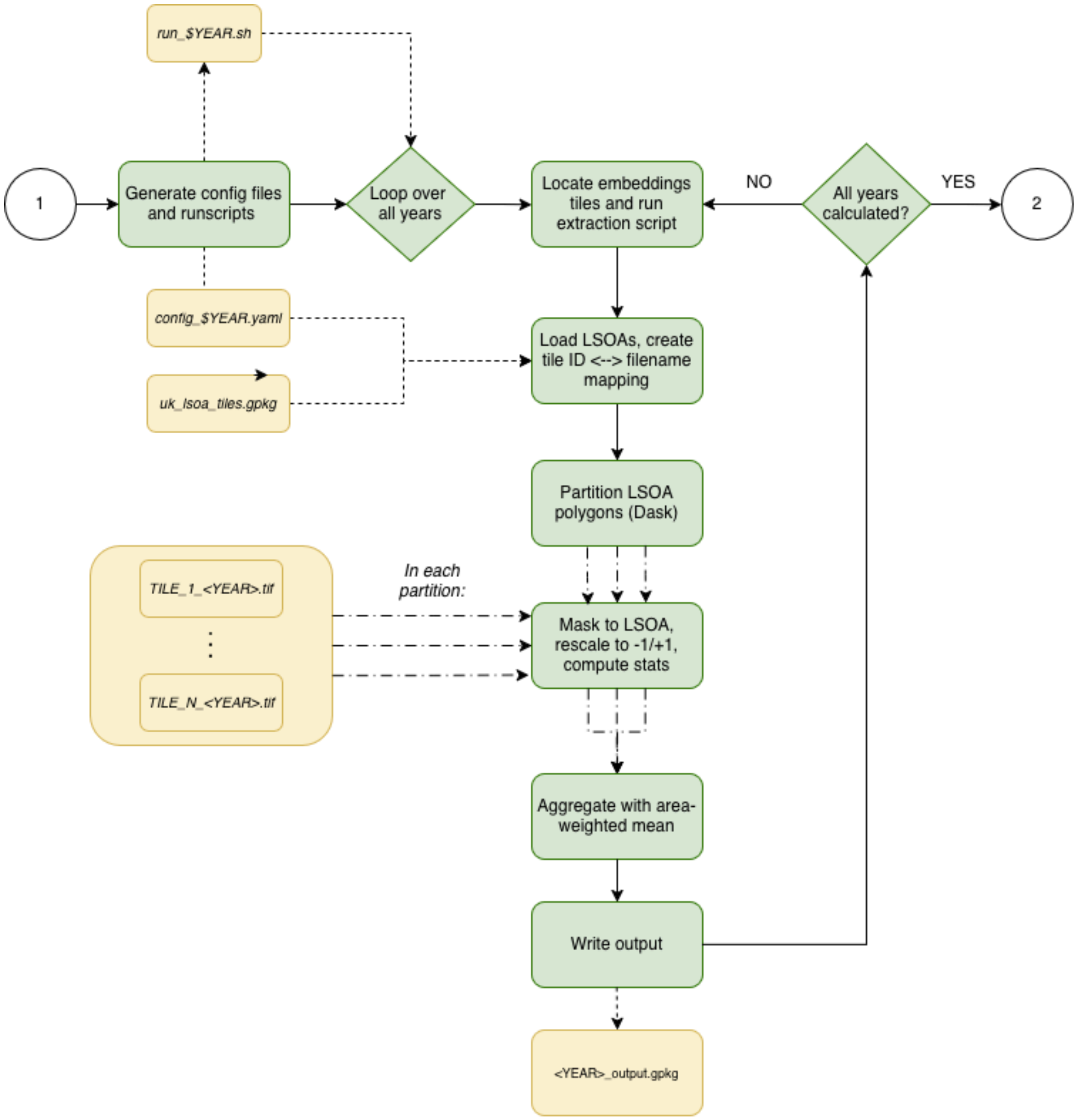

Figure 4: Conceptual flowchart, illustrating the input data, input/output operations, processing steps and output data in the LSOA aggregation pipeline

`embed_to_lsoa.py` calls routines from our `imago` toolkit repository to perform the actual LSOA calculation. In this step,
we create a mapping between tile IDs and filenames, and use these to load in the relevant tiles for each LSOA in the input LSOA GeoPackage file.
We then use `dask` to parallelise over the LSOAs in chunks, and within each `dask` worker, we use `rasterio` to perform zonal statistics to calculate
the mean embedding value for each dimension across all pixels that fall within the LSOA boundary (although see Limitations section below).

Once the mean embedding value for each pixel and band is calculated, we aggregate to calculate the mean embedding value for each dimension across the LSOA.
This averaging is calculated *per-tile*, and the sum of the pixels that overlap that tile is

saved. Then, the final LSOA-weighted average is the weighted average of
the per-tile averages, weighted by the number of pixels that overlap with the LSOA in each tile. This is necessary since some LSOAs will overlap with multiple tiles,
and we want to ensure that the final average is representative of the entire LSOA, rather than being biased towards the tile with the most pixels overlapping.

Finally, we write out the results to a file (default `GeoPackage` format, but optionally `parquet`), and re-run for the next year of interest.

Since we just rely on an input GeoPackage file, it is possible to perform this exact same aggregation for other geographies (e.g. MSOA), provided
that these files contain an appropriate tile <-> polygon mapping.

Limitations

The current pipeline uses `rasterio` for the zonal statistics calculation. This is faster than using libraries such as `exactextract`, but this has tradeoffs.
The pipeline treats all pixels that overlap with the LSOA as contributing the same amount to the pixel-weighted average. Thus, there is a small bias,
particularly for small LSOAs (i.e. those with high population density), since the proportion of pixels with partial overlap to pixels with full overlap is higher.
However, since the raster data is at very high (20m) resolution, this mitigates the impact of this bias significantly.

See the Validation section below for more details on the impact of this effect.

Performance

The datasets used for the LSOA aggregation step are accessed via Google Cloud Storage (GCS). Initially, these calculations were
done using Google Colab (for the Greater London area demo). However, due to compute limitations associated with Colab (particularly the limited number of CPUs available),
the LSOA aggregation step was run on a virtual machine (VM) with more resources (8 CPUs and 32 GB of memory). This number was
chosen as a balance between *good enough* performance and cost. We access the data from GCS via mounting the GCS bucket to the VM.
This does however have a performance impact, since I/O must navigate the additional overhead of accessing the data from GCS, which is not
as fast as reading the data from local disk.

The performance of the LSOA aggregation step is dependent on two main factors: the number of parallel `dask` workers used, and the
amount of memory given to each worker. For the initial release of the UK-wide dataset,

the pipeline was run on a VM with 8 workers
(CPUs) and 4 GB of memory per worker. In order to fit each dataset into memory, we use chunking to split the calculation into
groups of bands (e.g. 8 bands per chunk, so 8 chunks in total for the 64 bands). This allows us to run the processing
without spilling to disk, which heavily impacts performance.

Additionally, since we have lots of repeated I/O on tile metadata
(using `rasterio.open`), we use Python's `functools.lru_cache` to cache this data and avoid repeated reads; this improves overall
performance by a factor of ~25%.

On average, for each year, the LSOA aggregation step takes around 1.5 hours to run to completion. It is therefore possible
to run the entire LSOA extraction pipeline from 2017-2024 in a single day.

Performance can be improved by increasing the number of workers; however this is limited by the amount of memory available
on the machine. Possible performance enhancements include using a true windowed read to minimise memory pressure. This would
allow for more workers to be used, and also ease the need for band-by-band chunking, which will improve performance somewhat.

Technical Validation Report

LSOA-Level Satellite Embedding Data Product (2017-2024)

✓ VALIDATION PASSED - READY FOR PUBLICATION

Report Date: April 06, 2026

Data Coverage: 2017 - 2024 (8 years)

Geographic Level: LSOA (Lower Layer Super Output Area)

Total LSOAs per Year: 46,844

Embedding Dimensions: 64 features

1. Executive Summary

This report documents the validation of LSOA-level satellite embedding datasets spanning the years 2017 to 2024. The validation process consisted of two primary checks: (1) range verification to ensure all embedding values fall within the theoretical bounds of [-1, 1], and (2) random sampling validation comparing pipeline-generated data against direct Earth Engine extraction. Both validation procedures were successfully completed, and the datasets have been confirmed ready for publication on the Imago Data Catalogue.

Validation Overview

Validation Task	Description	Status
Range Check	Verify all embedding values are within [-1, 1]	✓ PASSED
Random Sampling Validation	Compare pipeline with EE extraction (100 samples)	✓ PASSED

2. Validation 1: Range Check [-1, 1]

The AlphaEarth embedding model produces values theoretically bounded within the range [-1, 1]. This validation systematically checked every numeric value across all 64 embedding dimensions for each year from 2017 to 2024.

2.1 Methodology

• Total values inspected per year: 46,844 LSOAs × 64 bands = 2,998,016 values
• Total values inspected across 8 years: ~24 million values
• Each value was checked against the [-1, 1] bounds

2.2 Results

Year	LSOAs	Dimensions	Total Values	Out-of-Range	Status
2017	46,844	64	2,998,016	0	✓
2018	46,844	64	2,998,016	0	✓
2019	46,844	64	2,998,016	0	✓
2020	46,844	64	2,998,016	0	✓
2021	46,844	64	2,998,016	0	✓
2022	46,844	64	2,998,016	0	✓
2023	46,844	64	2,998,016	0	✓
2024	46,844	64	2,998,016	0	✓

Key Finding: 100% compliance with the [-1, 1] range constraint across all years and all embedding dimensions.

2.3 Conclusion

The datasets satisfy the range validation requirement. All 24 million values across 8 years and 64 embedding dimensions are within the expected [-1, 1] bounds.

3. Validation 2: Random Sampling Against Earth Engine

3.1 Methodology

• Sample size: 100 randomly selected (LSOA, year) pairs
• Available pairs: 400 total (50 common LSOAs × 8 years)
• Comparison metrics: Mean difference, correlation, standard deviation
• Tolerance threshold: ±0.1 for mean difference
• Correlation threshold: > 0.5

3.2 Overall Statistics

Metric	Value
Total comparisons completed	100
Unique LSOAs sampled	42
Years covered	2017, 2018, 2019, 2020, 2021, 2022, 2023, 2024
Mean difference (Pipeline - EE)	-0.000049
Std deviation of differences	0.000698
Mean correlation	0.9973
Min correlation	0.9859
Max correlation	1.0000

3.3 Validation Pass Rates

Criterion	Threshold	Pass Rate
Mean values within tolerance	±0.1	100.0%
Correlation above threshold	> 0.5	100.0%

3.4 Systematic Bias Check

Test	Value	Result
t-statistic	-0.7085	
p-value	0.4803	PASS
Conclusion		No systematic bias detected

3.5 Yearly Breakdown

Year	Count	Mean Difference	Std Deviation
2017	11	-0.000027	0.000713
2018	13	-0.000176	0.000716
2019	13	0.000040	0.000887
2020	15	0.000082	0.000602

2021	10	0.000120	0.000648
2022	10	-0.000426	0.000538
2023	15	-0.000018	0.000821
2024	13	-0.000062	0.000605

4. Validation Visualizations

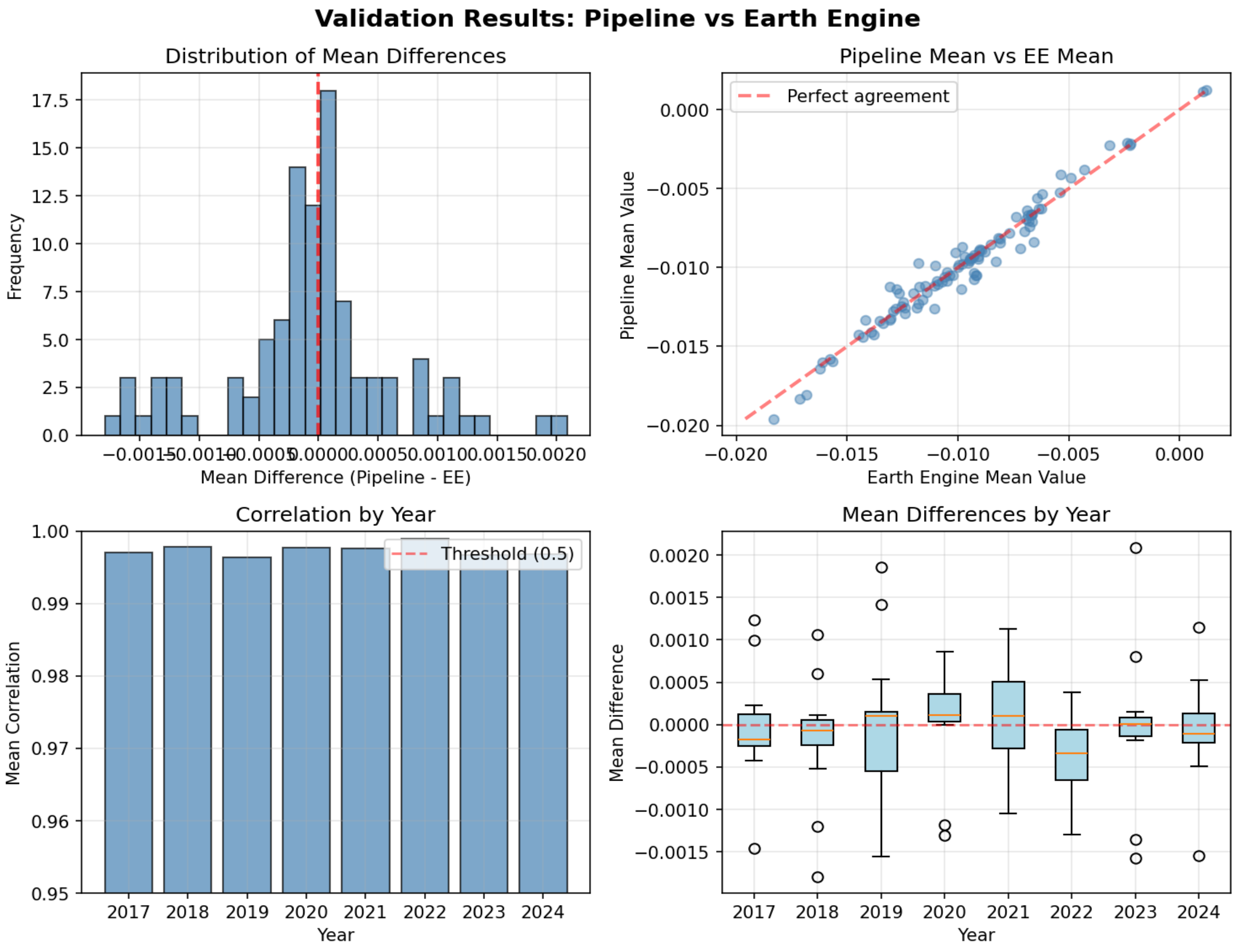

Figure 1: Validation visualizations showing distribution of mean differences, correlation between pipeline and EE data, and yearly performance.

5. Combined Validation Summary

Validation Criterion	Result	Recommendation
Range check [-1, 1]	✓ 100% compliant	Ready for publication
Statistical consistency with EE	✓ 100% pass rate	Ready for publication
Systematic bias	✓ None detected	Ready for publication
Year-to-year consistency	✓ Stable across all years	Ready for publication

5.1 Final Recommendation

The LSOA-level satellite embedding datasets for 2017-2024 are validated and recommended for publication on the Imago Data Catalogue.

Supporting Evidence:

• All 24 million values across all years and dimensions are within the theoretical [-1, 1] bounds
• Pipeline data shows near-perfect correlation (mean r = 0.997) with direct Earth Engine extraction
• Mean differences between pipeline and EE data are effectively zero (mean = -4.95e-05)
• No systematic bias detected across any year
• Validation covered 100 random (LSOA, year) pairs from 400 available combinations

Data Product Specifications (Confirmed):

Property	Value
Geographic level	LSOA (Lower Layer Super Output Area)
Temporal coverage	2017-2024 (8 years)
Embedding dimensions	64 features per LSOA
Number of LSOAs per year	46,844
Value range	[-1, 1] (verified)
Data format	GeoPackage (.gpkg)

6. Limitations and Future Work

6.1 Limitations of Current Validation

• The Earth Engine comparison was limited to 50 LSOAs per year that had complete data across both datasets
• The validation focuses on statistical consistency rather than exact pixel-level matching (different aggregation methods may produce slightly different values)

6.2 Recommendations for Future Validation

1. **Expand Earth Engine comparison** to include a larger set of LSOAs as data completeness improves
2. **Implement automated regression testing** for each new year of data added to the pipeline
3. **Add temporal consistency checks** to ensure year-over-year changes are plausible
4. **Develop outlier detection** for individual LSOA time series

Report generated: April 06, 2026 at 12:06:54 | Status: ✓ VALIDATED